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(54) **FITTINGS FOR JOINING LENGTHS OF PIPE AND PIPE ASSEMBLY FORMED USING SAME**

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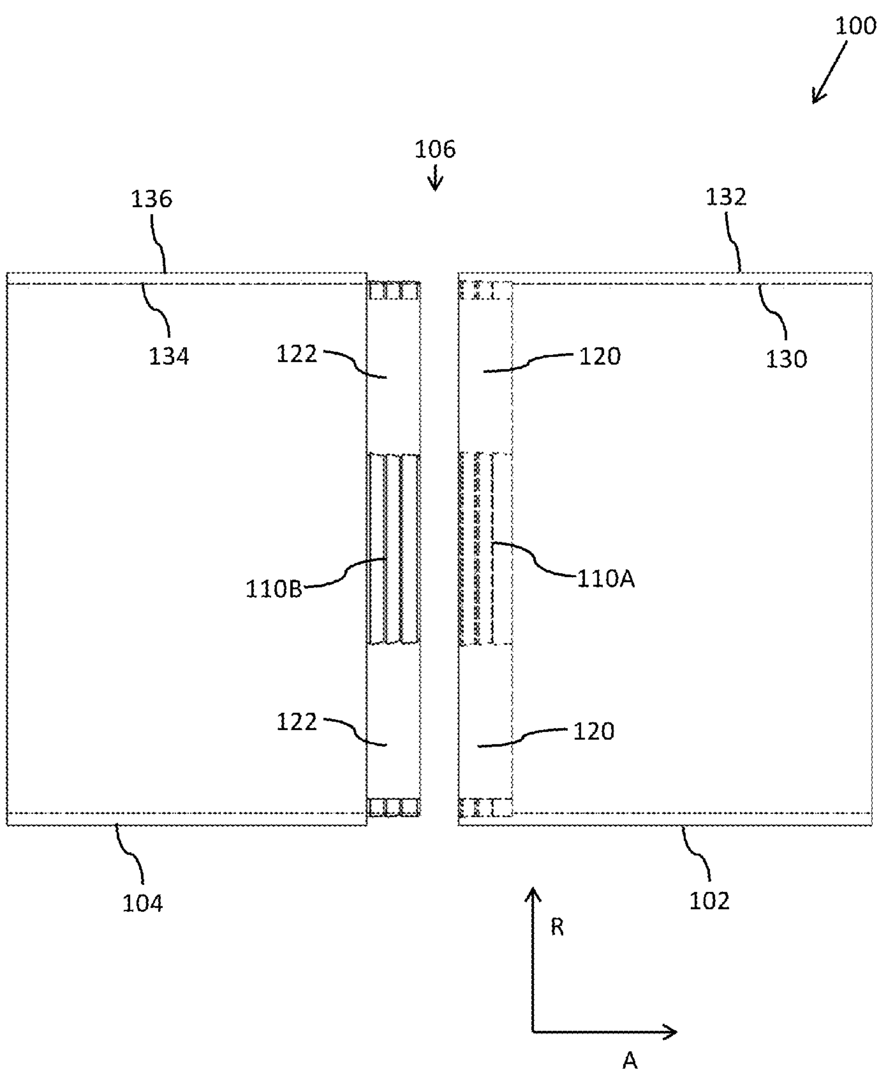
(57) **ABSTRACT**

A pipe assembly is provided, the pipe assembly comprising: a male fitting, comprising: an annular portion of pipe having a radially inner side and a radially outer side; at least one circumferentially-discontinuous section of teeth; and at least one circumferentially-discontinuous land; a female fitting, comprising: an annular portion of pipe having a radially inner side and a radially outer side; at least one circumferentially-discontinuous section of teeth; and at least one circumferentially-discontinuous groove.



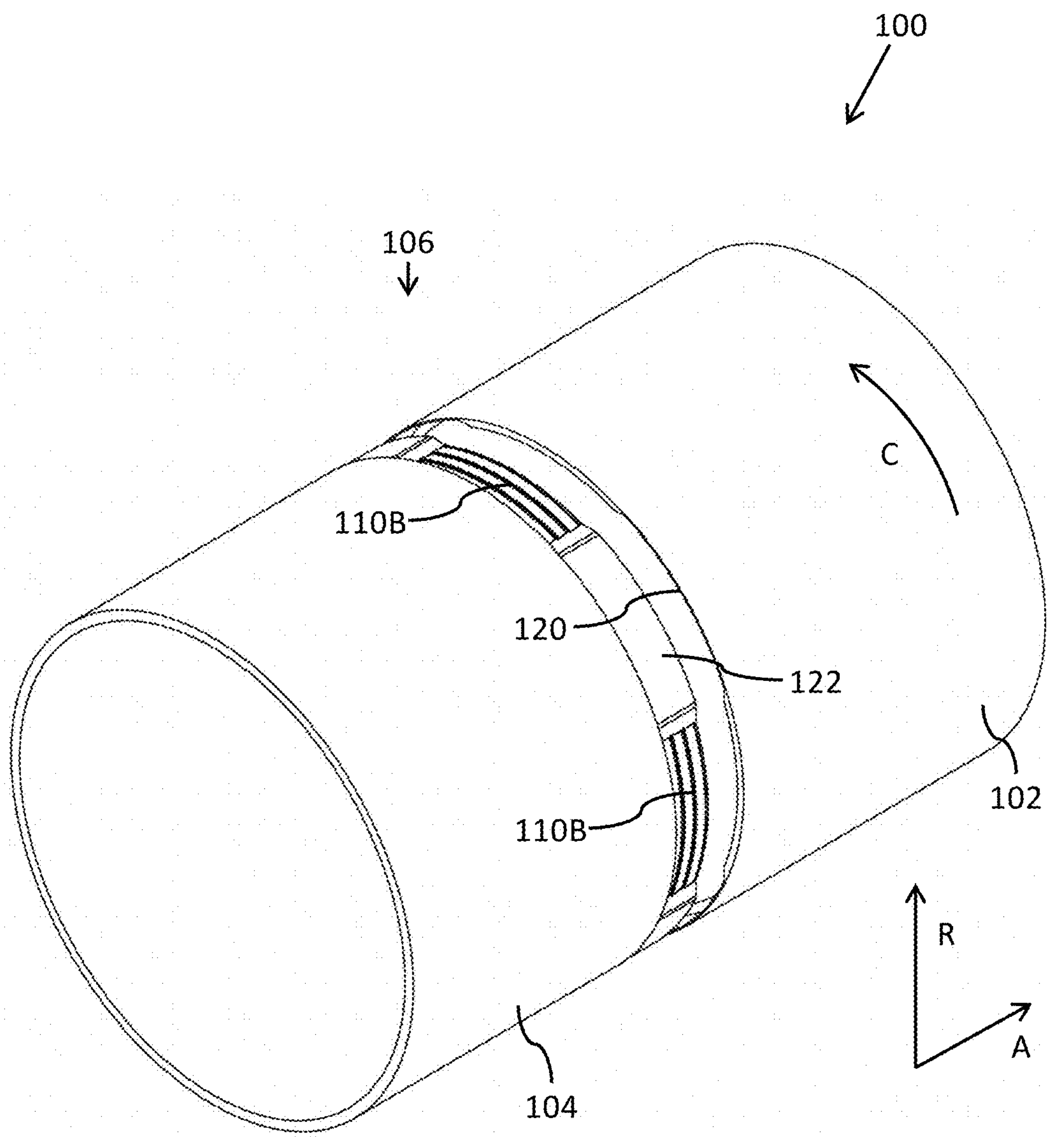


FIG. 1A

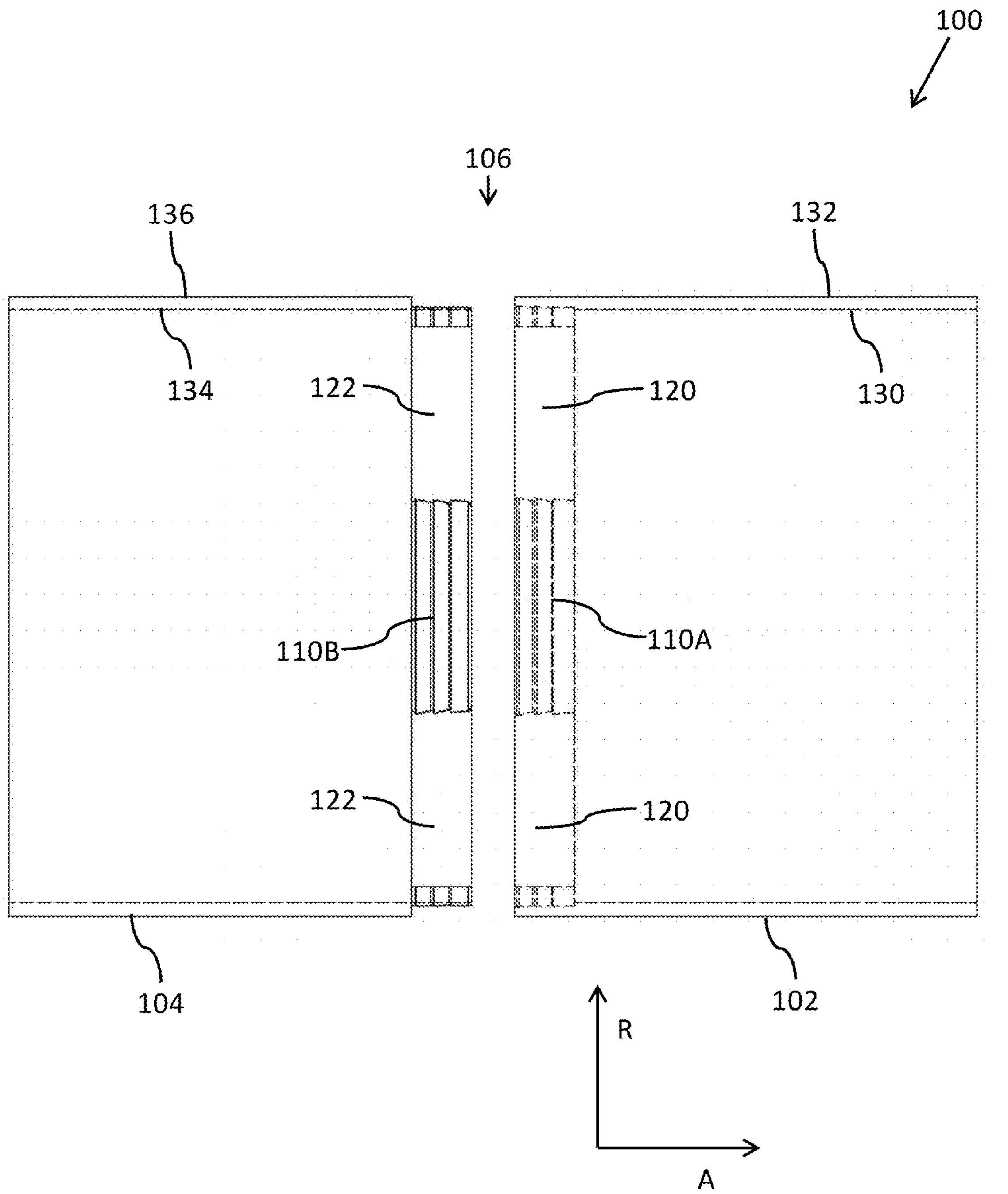


FIG. 1B

FITTINGS FOR JOINING LENGTHS OF PIPE AND PIPE ASSEMBLY FORMED USING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from U.S. Provisional Patent Application No. 63/562,622 filed on Mar. 7, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Creating a pipeline, whether horizontally or vertically, and whether underground or above ground, one typically assembles consecutive lengths of pipe one after the other.

[0003] Traditionally, individual lengths of pipe used in forming a pipeline are arranged in a line, end-to-end, with each being welded to the next as individual lengths of pipe are added to the pipeline. However, welding the individual lengths of pipe to one another is a time-consuming process involving additional danger, high heat, skilled welding personnel, time to allow welds to cool, and possible imperfections in the finished product.

[0004] Accordingly, what is needed is a pipe connection that requires less time and work to assemble, while being able to withstand the compressive and bending stresses that the pipes will experience both during installation and operation in the field.

SUMMARY

[0005] In one aspect, a pipe assembly is provided, the pipe assembly comprising: a male fitting, comprising: an annular portion of pipe having a radially inner side and a radially outer side; at least one circumferentially-discontinuous section of teeth; and at least one circumferentially-discontinuous land; a female fitting, comprising: an annular portion of pipe having a radially inner side and a radially outer side; at least one circumferentially-discontinuous section of teeth; and at least one circumferentially-discontinuous groove.

BRIEF DESCRIPTION OF THE FIGURES

[0006] The accompanying figures, which are incorporated in and constitute a part of the specification, illustrate various example embodiments, and are used merely to illustrate various example embodiments. In the figures, like elements bear like reference numerals.

[0007] FIG. **1**A illustrates a perspective view of a pipe assembly **100**, including a female fitting **102** and a male fitting **104**.

[0008] FIG. **1**B illustrates a sectional view of a pipe assembly **100**, including a female fitting **102** and a male fitting **104**.

DETAILED DESCRIPTION

[0009] As used throughout this application, the radial direction refers to direction R as illustrated in the drawings. Radially outer or radially outwardly refers to the direction R as the radius is increased, whereas radially inner or radially inwardly refers to the direction R as the radius is decreased. As used throughout this application, the axial direction refers to that extending along or parallel to direction A. The axial direction A is intended to extend along the longitudinal length of a pipe assembly. As used throughout this application, the circumferential direction refers to direction C, which extends about the circumference of the pipe assembly in a plane orthogonal to the axial direction A, and parallel to the radial direction R.

[0010] FIGS. **1**A and **1**B illustrate a pipe assembly **100**, including a female fitting **102** and a male fitting **104** at a junction **106**.

[0011] Female fitting **102** may include an annular portion of pipe having a radially inner side **130** and a radially outer side **132**. Radially inner side **130** of female fitting **102** includes teeth **110**A and grooves **120** at its distal end.

[0012] Male fitting **104** may include an annular portion of pipe having a radially inner side **134** and a radially outer side **136**. Radially outer side **136** of male fitting **104** includes circumferentially-discontinuous sections of teeth **110**B and lands **122** at its distal end, corresponding to those of female fitting **102**, as will be further described below.

[0013] Female fitting **102** may include at least one circumferentially-discontinuous section of teeth **110**A. Female fitting **102** may include at least one circumferentially-discontinuous groove **120**. Female fitting **102** may include a plurality of circumferentially-discontinuous sections of teeth **110**A alternating with a plurality of circumferentially-discontinuous grooves **120**. In one aspect, female fitting **102** includes four circumferentially-discontinuous sections of teeth **110**A equally distributed about the circumference of female fitting **102**, centered 90 degrees apart from one another, and four circumferentially-discontinuous grooves **120** equally distributed about the circumference of female fitting **102**, centered 90 degrees apart from one another, wherein each of the grooves **120** are oriented between two sections of teeth **110**A and wherein each section of teeth **110**A are oriented between two grooves **120**. In another aspect, female fitting **102** includes three circumferentially-discontinuous sections of teeth **110**A equally distributed about the circumference of female fitting **102**, centered 120 degrees apart from one another, and three circumferentially-discontinuous grooves **120** equally distributed about the circumference of female fitting **102**, centered 120 degrees apart from one another, wherein each of the grooves **120** are oriented between two sections of teeth **110**A and wherein each section of teeth **110**A are oriented between two grooves **120**. In another aspect, female fitting **102** includes three, four, or five circumferentially-discontinuous sections of teeth **110**A equally distributed about the circumference of female fitting **102**, and three, four, or five circumferentially-discontinuous grooves **120** equally distributed about the circumference of female fitting **102**, wherein each of the grooves **120** are oriented between two sections of teeth **110**A and wherein each section of teeth **110**A are oriented between two grooves **120**.

[0014] Male fitting **104** may include at least one circumferentially-discontinuous section of teeth **110**B. Male fitting **104** may include at least one circumferentially-discontinuous land **122**. Male fitting **104** may include a plurality of circumferentially-discontinuous sections of teeth **110**B alternating with a plurality of circumferentially-discontinuous lands **122**. In one aspect, male fitting **104** includes four circumferentially-discontinuous sections of teeth **110**B equally distributed about the circumference of male fitting **104**, centered 90 degrees apart from one another, and four circumferentially-discontinuous lands **122** equally distributed about the circumference of male fitting **104**, centered 90 degrees apart from one another, wherein each of the lands **122** are oriented between two sections of teeth **110**B and wherein each section of teeth **110**B are oriented between two lands **122**. In another aspect, male fitting **104** includes three circumferentially-discontinuous sections of teeth **110**B equally distributed about the circumference of male fitting **104**, centered 120 degrees apart from one another, and three circumferentially-discontinuous lands **122** equally distributed about the circumference of male fitting **104**, centered 120 degrees apart from one another, wherein each of the lands **122** are oriented between two sections of teeth **110**B and wherein each section of teeth **110**B are oriented between two lands **122**. In another aspect, male fitting **104** includes three, four, or five circumferentially-discontinuous sections of teeth **110**B equally distributed about the circumference of male fitting **104**, and three, four, or five circumferentially-discontinuous lands **122** equally distributed about the circumference of male fitting **104**, wherein each of the lands **122** are oriented between two sections of teeth **110**B and wherein each section of teeth **110**B are oriented between two lands **122**.

[0015] Regardless of the number of sections of teeth **110**A, **110**B, grooves **120**, and lands **122**, female fitting **102** and male fitting **104** have the same number of sections of teeth **110**A, **110**B, and the same number of grooves **120** and lands **122**. For example, if female fitting **102** includes four sections of teeth **110**A, then male fitting **104** includes four sections of teeth **110**B, and if female fitting **102** includes three grooves **120**, then male fitting **104** includes three lands **122**. Stated simply, each section of teeth **110**A in assembly **100** has a corresponding section of teeth **110**B, and each groove **120** in assembly **100** has a corresponding land **122**.

[0016] At least one of female fitting **102** and male fitting **104** may be formed independently of a length of pipe, and may welded or otherwise mechanically fixed to the length of pipe at a proximal end of female fitting **102** and/or male fitting **104**. This welding or mechanical attachment of female fitting **102** and/or male fitting **104** to a length of pipe may be performed away from the site of installation of the pipe, or on the site of installation of the pipe.

[0017] Alternatively, at least one of female fitting **102** and male fitting **104** may be machined directly into the ends of a length of pipe. The machining of female fitting **102** and male fitting **104** may be performed using known machining techniques.

[0018] While the drawings focus on the fittings fixing lengths of pipe to one another, rather than the lengths of pipe themselves, it is understood that the fittings are oriented at the ends of a length of pipe. In practice, a single length of pipe may include female fitting **102** at a first end, and male fitting **104** at a second end. In this manner, a series of like lengths of pipe may be fit together, with the female end of a first length of pipe engaging the male end of a second length of pipe, while the male end of the first length of pipe engages the female end of a third length of pipe.

[0019] Alternatively, a single length of pipe may include female fitting **102** at both its first ends and second ends, or may include male fitting **104** at both its first ends and second ends. In this manner, the independent lengths of pipe would be either "male" lengths (with male fitting **104** at each end), or "female" lengths (with female fitting **102** at each end). Assembling a pipeline using these lengths of pipe would require alternating between male lengths and female lengths during assembly of the pipeline.

[0020] Female fitting **102** and male fitting **104** may be made of any of a variety of materials, including without limitation, a metal (such as steel), an alloy, a polymer, a composite, and the like. In order for female fitting **102** and male fitting **104** to deflect enough relative to one another during the press-fitting of the two together, the material may need appropriate elastic qualities.

[0021] Lengths of pipe used in association with female fitting **102** and male fitting **104** may include any of a variety of lengths convenient for manufacture, transport, and assembly.

[0022] Diameters of lengths of pipe used in association with female fitting **102** and male fitting **104**, and thus the diameters of female fitting **102** and male fitting **104**, can be any of a variety of diameters. For example, the pipe and fittings may have outer diameters between about 6 in. (15 cm) and 127 in. (323 cm). The pipe and fittings may have outer diameters outside of this range as well.

[0023] Female fitting **102** includes teeth **110**A. Male fitting **104** includes teeth **110**B. Teeth **110**A and **110**B are configured (i.e., sized and shaped) to engage one another upon insertion of male fitting **104** into female fitting **102** to prevent movement of female fitting **102** relative to male fitting **104** in axial direction A.

[0024] Teeth **110**A and **110**B may include at least one standard tooth extending circumferentially along the circumferentially-discontinuous sections of teeth **110**A, **110**B. The at least one standard tooth includes an engagement surface parallel to the radial direction R.

[0025] Teeth **110**A and **110**B may include at least one biased tooth extending circumferentially along the circumferentially-discontinuous sections of teeth **110**A, **110**B. The at least one biased tooth includes and engagement surface biased at an angle relative to the radial direction R. The angle may be any of a variety of angles, including for example, 22 degrees. The angle may be between 20 degrees and 24 degrees. The angle may be between 18 degrees and 26 degrees. The angle may be between 16 degrees and 28 degrees. The angle may be between 14 degrees and 30 degrees. The angle may be between 12 degrees and 32 degrees.

[0026] Assembly **100** may include two, three, or four teeth **110**A and **110**B. Assembly **100** may include an equal number of teeth **110**A as teeth **110**B in a given section of teeth.

[0027] The biased engagement surfaces may be biased toward the proximal direction of the corresponding female fitting **102** and male fitting **104**. That is, the biased tooth of female fitting **102** is biased such that the biased engagement surface extends in the proximal direction of female fitting **102** as the engagement surface extends radially away from the body of female fitting **102**. The biased tooth of male fitting **104** is biased such that the biased engagement surface extends in a proximal direction of male fitting **104** as the engagement surface extends radially away from the body of male fitting **104**.

[0028] Female fitting **102** includes grooves **120**. Male fitting **104** includes lands **122**. Grooves **120** and lands **122** are configured (i.e., sized and shaped) to engage one another upon insertion of male fitting **104** into female fitting **102** to prevent rotation of female fitting **102** relative to male fitting **104** in circumferential direction C about an axis parallel to axial direction A. Lands **122** are sized, shaped, and spaced to fit into grooves **120**, while grooves **120** are sized, shaped, and spaced to accept lands **122**.

[0029] Grooves **120** may be machined into the thickenss of female fitting **102** from radially inner side **130**. Lands **122** may be machined into the thickness of male fitting **104** from radially outer side **136**. Teeth **110**A may be machined into the thickenss of female fitting **102** from radially inner side **130**. Teeth **110**B may be machined into the thickness of male fitting **104** from radially outer side **136**.

[0030] To the extent that the term "includes" or "including" is used in the specification or the claims, it is intended to be inclusive in a manner similar to the term "comprising" as that term is interpreted when employed as a transitional word in a claim. Furthermore, to the extent that the term "or" is employed (e.g., A or B) it is intended to mean "A or B or both." When the applicants intend to indicate "only A or B but not both" then the term "only A or B but not both" will be employed. Thus, use of the term "or" herein is the inclusive, and not the exclusive use. See Bryan A. Garner, A Dictionary of Modern Legal Usage 624 (2d. Ed. 1995). Also, to the extent that the terms "in" or "into" are used in the specification or the claims, it is intended to additionally mean "on" or "onto." To the extent that the term "substantially" is used in the specification or the claims, it is intended to take into consideration the degree of precision available or prudent in manufacturing. To the extent that the term "selectively" is used in the specification or the claims, it is intended to refer to a condition of a component wherein a user of the apparatus may activate or deactivate the feature or function of the component as is necessary or desired in use of the apparatus. To the extent that the term "operatively connected" is used in the specification or the claims, it is intended to mean that the identified components are connected in a way to perform a designated function. As used in the specification and the claims, the singular forms "a," "an," and "the" include the plural. Finally, where the term "about" is used in conjunction with a number, it is intended to include ±10% of the number. In other words, "about 10" may mean from 9 to 11.

[0031] As stated above, while the present application has been illustrated by the description of embodiments thereof, and while the embodiments have been described in considerable detail, it is not the intention of the applicants to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art, having the benefit of the present application. Therefore, the application, in its broader aspects, is not limited to the specific details, illustrative examples shown, or any apparatus referred to. Departures may be made from such details, examples, and apparatuses without departing from the spirit or scope of the general inventive concept.

1. A pipe assembly, comprising:
a male fitting, comprising:
an annular portion of pipe having a radially inner side and a radially outer side;
at least one circumferentially-discontinuous section of teeth; and
at least one circumferentially-discontinuous land;
a female fitting, comprising:
an annular portion of pipe having a radially inner side and a radially outer side;
at least one circumferentially-discontinuous section of teeth; and
at least one circumferentially-discontinuous groove.

2. The pipe assembly of claim **1**, wherein the male fitting and the female fitting include the same number of circumferentially-discontinuous sections of teeth.

3. The pipe assembly of claim **1**, wherein the male fitting includes the same number of circumferentially-discontinuous lands as the female fitting includes circumferentially-discontinuous grooves.

4. The pipe assembly of claim **1**, wherein the male fitting includes a section of teeth having at least one standard tooth extending circumferentially along the circumferentially-discontinuous section of teeth, wherein the standard tooth includes an engagement surface parallel to a radial direction of the pipe assembly.

5. The pipe assembly of claim **1**, wherein the male fitting includes a section of teeth having at least one biased tooth extending circumferentially along the circumferentially-discontinous section of teeth, wherein the biased tooth includes an engagement surface biased at an angle relative to a radial direction of the pipe assembly.

6. The pipe assembly of claim **1**, wherein the female fitting includes a section of teeth having at least one standard tooth extending circumferentially along the circumferentially-discontinuous section of teeth, wherein the standard tooth includes an engagement surface parallel to a radial direction of the pipe assembly.

7. The pipe assembly of claim **1**, wherein the female fitting includes a section of teeth having at least one biased tooth extending circumferentially along the circumferentially-discontinuous section of teeth, wherein the biased tooth includes an engagement surface biased at an angle relative to a radial direction of the pipe assembly.

8. The pipe assembly of claim **1**, wherein each male fitting circumferentially-discontinuous section of teeth engages each female fitting circumferentially-discontinuous section of teeth.

9. The pipe assembly of claim **1**, wherein each male fitting circumferentially-discontinuous land engages each female fitting circumferentially-discontinuous groove.

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